

INVESTMENTS

7th
Edition

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Bond Pricing between Coupon Dates

Equation 14.2 for bond prices assumes that the next coupon payment is in precisely one payment period, either a year for an annual payment bond or 6 months for a semiannual payment bond. But you probably want to be able to price bonds all 365 days of the year, not just on the one or two dates each year that it makes a coupon payment!

In principle, the fact that the bond is between coupon dates does not affect the pricing problem. The procedure is always the same: compute the present value of each remaining payment and sum up. But if you are between coupon dates, there will be fractional periods remaining until each payment, and this does complicate the arithmetic computations.

Fortunately, bond pricing functions are included in most spreadsheet programs such as Excel. The spreadsheet allows you to enter today's date as well as the maturity date of the bond, and so can provide prices for bonds at any date. The nearby box shows you how.

As we pointed out earlier, bond prices are typically quoted net of accrued interest. These prices, which appear in the financial press, are called *flat prices*. The actual *invoice price* that a buyer pays for the bond includes accrued interest. Thus,

$$\text{Invoice price} = \text{Flat price} + \text{Accrued interest}$$

When a bond pays its coupon, flat price equals invoice price, since at that moment accrued interest reverts to zero. However, this will be the exceptional case, not the rule.

The Excel pricing function provides the flat price of the bond. To find the invoice price, we need to add accrued interest. Fortunately, Excel also provides functions that count the days since the last coupon payment date and thus can be used to compute accrued interest. The nearby box also illustrates how to use these functions. The box provides examples using bonds that have just paid a coupon and so have zero accrued interest, as well as a bond that is between coupon dates.

14.3 BOND YIELDS

We have noted that the current yield of a bond measures only the cash income provided by the bond as a percentage of bond price and ignores any prospective capital gains or losses. We would like a measure of rate of return that accounts for both current income and the price increase or decrease over the bond's life. The yield to maturity is the standard measure of the total rate of return. However, it is far from perfect, and we will explore several variations of this measure.

Yield to Maturity

In practice, an investor considering the purchase of a bond is not quoted a promised rate of return. Instead, the investor must use the bond price, maturity date, and coupon payments to infer the return offered by the bond over its life. The **yield to maturity (YTM)** is defined as the interest rate that makes the present value of a bond's payments equal to its price. This interest rate is often viewed as a measure of the average rate of return that will be earned on a bond if it is bought now and held until maturity. To calculate the yield to maturity, we solve the bond price equation for the interest rate given the bond's price.

Excel

Bond P

Excel and built-in functions. They typically use the `PRICE` function to calculate the price of the bond (c

`= PRICE(set rate, yield of par valu`

For the 4.25% in Figure 14.1, the following spreadsheet

`= PRICE(DAT`

The DATE function for the settlement date (year, month, day) is 2006, when the January 15, 2006 coupon is due. Notice that the dates are expressed as serial numbers (e.g., 365 for one year, 182.5 for six months, etc.), and the rate is expressed as a percent of par value. The result is the price of the bond to par value. On the next page, we discussed shortly.

Excel

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1	
2	
3	
4	Settlement date
5	Maturity date
6	Annual coupon rate
7	Yield to maturity
8	Redemption value
9	Coupon payment
10	
11	
12	Flat price (% of par)
13	Days since last coupon
14	Days in coupon period
15	Accrued interest
16	Invoice price

Bond Pricing

Excel and most other spreadsheet programs provide built-in functions to compute bond prices and yields. They typically ask you to input both the date you buy the bond (called the *settlement date*) and the maturity date of the bond. The Excel function for bond price is

= PRICE(settlement date, maturity date, annual coupon rate, yield to maturity, redemption value as percent of par value, number of coupon payments per year)

For the 4.25% coupon 2011 maturity bond highlighted in Figure 14.1, we would enter the values in the following spreadsheet. Alternatively, we could simply enter the following function in Excel:

= PRICE(DATE(2006,1,15), DATE(2011,1,15), .0425, .0427, 100, 2)

The DATE function in Excel, which we use for both the settlement and maturity date, uses the format DATE(year,month,day). The first date is January 15, 2006, when the bond is purchased, and the second is January 15, 2011, when it matures. Most bonds pay coupons on the 15th of the month.

Notice that the coupon rate and yield to maturity are expressed as decimals, not percentages. In most cases, redemption value is 100 (i.e., 100% of par value), and the resulting price similarly is expressed as a percent of par value. Occasionally, however, you may encounter bonds that pay off at a premium or discount to par value. One example would be callable bonds, discussed shortly.

The value of the bond returned by the pricing function is 99.911 (cell B12), which matches the price reported in *The Wall Street Journal*

to the nearest 32nd of a point. This bond has just paid a coupon. In other words, the settlement date is precisely at the beginning of the coupon period, so no adjustment for accrued interest is necessary.

To illustrate the procedure for bonds between coupon payments, consider the 4% coupon November 2012 bond, also highlighted in Figure 14.1. Using the entries in column D of the spreadsheet, we find in cell D12 that the (flat) price of the bond is 98.235, which matches the price given in *The Wall Street Journal* except for a few cents rounding error.

What about the bond's invoice price? Rows 13 through 16 make the necessary adjustments. The function described in cell C13 counts the days since the last coupon. This day count is based on the bond's settlement date, maturity date, coupon period (1 = annual; 2 = semiannual), and day count convention (choice 1 uses actual days). The function described in cell C14 counts the total days in each coupon payment period. Therefore, the entries for accrued interest in row 15 are the semiannual coupon multiplied by the fraction of a coupon period that has elapsed since the last payment. Finally, the invoice prices in row 16 are the sum of flat price plus accrued interest.

As a final example, suppose you wish to find the price of the bond in Example 14.2. It is a 30-year maturity bond with a coupon rate of 8% (paid semiannually). The market interest rate given in the latter part of the example is 10%. However, you are not given a specific settlement or maturity date. You can still use the PRICE function to value the bond. Simply choose an *arbitrary* settlement date (January 1, 2000, is convenient) and let the maturity date be 30 years hence. The appropriate inputs appear in column F of the spreadsheet, with the resulting price, 81.071% of face value, appearing in cell F16.

Excel

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	A	B	C	D	E	F	G
1	4.250% coupon bond,			4% coupon bond,		8% coupon bond,	
2	maturing Jan 2011		Formula in column B	maturing Nov 2012		30-year maturity	
3							
4	Settlement date	1/15/2006	= DATE (2006, 1, 15)	1/15/2006		1/1/2000	
5	Maturity date	1/15/2011	= DATE (2011, 1, 15)	11/15/2012		1/1/2030	
6	Annual coupon rate	0.0425		0.04		0.08	
7	Yield to maturity	0.0427		0.043		0.1	
8	Redemption value (% of face value)	100		100		100	
9	Coupon payments per year	2		2		2	
10							
11							
12	Flat price (% of par)	99.911	=PRICE(B4,B5,B6,B7,B8,B9)	98.235		81.071	
13	Days since last coupon	0	=COUPDAYBS(B4,B5,2,1)	61		0	
14	Days in coupon period	181	=COUPDAYS(B4,B5,2,1)	181		182	
15	Accrued interest	0	=B13/B14*B6*100/2	0.0674		0	
16	Invoice price	99.911	=B12+B15	98.909		81.071	

EXAMPLE 14.3 Yield to Maturity

Suppose an 8% coupon, 30-year bond is selling at \$1,276.76. What average rate of return would be earned by an investor purchasing the bond at this price? We find the interest rate at which the present value of the remaining 60 semiannual payments equals the bond price. This is the rate consistent with the observed price of the bond. Therefore, we solve for r in the following equation:

$$\$1,276.76 = \sum_{t=1}^{60} \frac{\$40}{(1+r)^t} + \frac{\$1,000}{(1+r)^{60}}$$

or, equivalently,

$$1,276.76 = 40 \times \text{Annuity factor}(r, 60) + 1,000 \times \text{PV factor}(r, 60)$$

These equations have only one unknown variable, the interest rate, r . You can use a financial calculator or spreadsheet to confirm that the solution is $r = .03$, or 3%, per half-year.⁸ This is considered the bond's yield to maturity.

The financial press reports yields on an annualized basis, and annualizes the bond's semiannual yield using simple interest techniques, resulting in an annual percentage rate, or APR. Yields annualized using simple interest are also called "bond equivalent yields." Therefore, the semiannual yield would be doubled and reported in the newspaper as a bond equivalent yield of 6%. The *effective* annual yield of the bond, however, accounts for compound interest. If one earns 3% interest every 6 months, then after 1 year, each dollar invested grows with interest to $\$1 \times (1.03)^2 = \1.0609 , and the effective annual interest rate on the bond is 6.09%.

Excel also provides a function for yield to maturity that is especially useful in between coupon dates. It is

=YIELD(settlement date, maturity date, annual coupon rate, bond price, redemption value as percent of par value, number of coupon payments per year)

The bond price used in the function should be the reported flat price, without accrued interest. For example, to find the yield to maturity of the bond in Example 14.3, we would use column E of Spreadsheet 14.1. If the coupons were paid only annually, we would change the entry for payments per year to 1 (see cell D8), and the yield would fall slightly to 5.99%.

The bond's yield to maturity is the internal rate of return on an investment in the bond. The yield to maturity can be interpreted as the compound rate of return over the life of the bond under the assumption that all bond coupons can be reinvested at that yield.⁹ Yield to maturity is widely accepted as a proxy for average return.

Yield to maturity differs from the **current yield** of a bond, which is the bond's annual coupon payment divided by the bond price. For example, for the 8%, 30-year bond currently selling at \$1,276.76, the current yield would be $\$80/\$1,276.76 = .0627$, or 6.27%, per

⁸On your financial calculator, you would enter the following inputs: $n = 60$ periods; $PV = -1,276.76$; $FV = 1,000$; $PMT = 40$; then you would compute the interest rate (COMP i or CPT i). Notice that we enter the present value, or PV, of the bond as *minus* \$1,000. Again, this is because most calculators treat the initial purchase price of the bond as a cash outflow. Without a financial calculator or spreadsheet, you still could solve the equation, but you would need to use a trial-and-error approach.

⁹If the reinvestment rate does not equal the bond's yield to maturity, the compound rate of return will differ from YTM. This is demonstrated in Examples 14.5 and 14.6.

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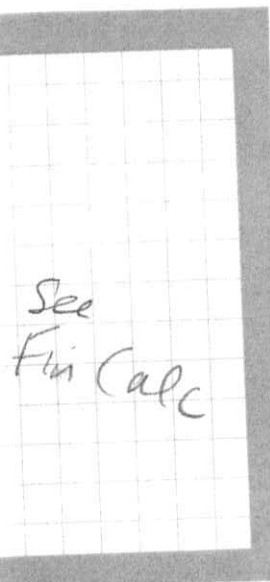
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1	Annual coupons			Semiannual coupons	
2					
3	Settlement date	1/1/2000		1/1/2000	
4	Maturity date	1/1/2030		1/1/2030	
5	Annual coupon rate	0.08		0.08	
6	Bond price	127.676		127.676	
7	Redemption value (% of face value)	100		100	
8	Coupon payments per year	2		1	
9					
10	Yield to maturity (decimal)	0.0600		0.0599	
11					
12	The formula entered here is : =YIELD(B3,B4,B5,B6,B7,B8)				

eXcel

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SPREADSHEET 14.1

Finding yield to maturity in Excel

year. In contrast, recall that the effective annual yield to maturity is 6.09%. For this bond, which is selling at a premium over par value (\$1,276 rather than \$1,000), the coupon rate (8%) exceeds the current yield (6.27%), which exceeds the yield to maturity (6.09%). The coupon rate exceeds current yield because the coupon rate divides the coupon payments by par value (\$1,000) rather than by the bond price (\$1,276). In turn, the current yield exceeds yield to maturity because the yield to maturity accounts for the built-in capital loss on the bond; the bond bought today for \$1,276 will eventually fall in value to \$1,000 at maturity.

Example 14.3 illustrates a general rule: for **premium bonds** (bonds selling above par value), coupon rate is greater than current yield, which in turn is greater than yield to maturity. For **discount bonds** (bonds selling below par value), these relationships are reversed (see Concept Check 3).

It is common to hear people talking loosely about the yield on a bond. In these cases, they almost always are referring to the yield to maturity.

CONCEPT CHECK

3

What will be the relationship among coupon rate, current yield, and yield to maturity for bonds selling at discounts from par? Illustrate using the 8% (semiannual payment) coupon bond, assuming it is selling at a yield to maturity of 10%.

Yield to Call

Yield to maturity is calculated on the assumption that the bond will be held until maturity. What if the bond is callable, however, and may be retired prior to the maturity date? How should we measure average rate of return for bonds subject to a call provision?

Figure 14.4 illustrates the risk of call to the bondholder. The colored line is the value at various market interest rates of a “straight” (i.e., noncallable) bond with par value \$1,000, an 8% coupon rate, and a 30-year time to maturity. If interest rates fall, the bond price, which equals the present value of the promised payments, can rise substantially.

Now consider a bond that has the same coupon rate and maturity date but is callable at 110% of par value, or \$1,100. When interest rates fall, the present value of the bond’s scheduled payments rises, but the call provision allows the issuer to repurchase the bond at the call price. If the call price is less than the present value of the scheduled payments, the issuer may call the bond back from the bondholder.

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TEXAS
INSTRUMENTS

Bonds¹

Price (given yield) with one coupon period or less to redemption:

$$PRI = \left[\frac{RV + \frac{100 \times R}{M}}{1 + \left(\frac{DSR}{E} \times \frac{Y}{M} \right)} \right] - \left[\frac{A}{E} \times \frac{100 \times R}{M} \right]$$

where: PRI = dollar price per \$100 par value

RV = redemption value of the security per \$100 par value ($RV = 100$ except in those cases where call or put features must be considered)

R = annual interest rate (as a decimal; $CPN \div 100$)

M = number of coupon periods per year standard for the particular security involved (set to 1 or 2 in Bond worksheet)

DSR = number of days from settlement date to redemption date (maturity date, call date, put date, etc.)

E = number of days in coupon period in which the settlement date falls

Y = annual yield (as a decimal) on investment with security held to redemption ($YLD \div 100$)

A = number of days from beginning of coupon period to settlement date (accrued days)

Note: The first term computes present value of the redemption amount, including interest, based on the yield for the invested period. The second term computes the accrued interest agreed to be paid to the seller.

Yield (given price) with one coupon period or less to redemption:

$$Y = \left[\frac{\left(\frac{RV + R}{100 + \frac{R}{M}} \right) - \left(\frac{PRI}{100} + \left(\frac{A}{E} \times \frac{R}{M} \right) \right)}{\frac{PRI}{100} + \left(\frac{A}{E} \times \frac{R}{M} \right)} \right] \times \left[\frac{M \times E}{DSR} \right]$$

1. Source for bond formulas (except duration): Lynch, Jr., and Jan H. Mayle, Standard Securities Calculation Methods, New York: Securities Industry Association, 1986.

Price (given yield) with more than one coupon period to redemption:

$$PRI = \left[\frac{RV}{\left(1 + \frac{Y}{M}\right)^{N-1 + \frac{DSC}{E}}} \right] + \sum_{K=1}^N \left[\frac{100 \times \frac{R}{M}}{\left(1 + \frac{Y}{M}\right)^{K-1 + \frac{DSC}{E}}} \right] - \left[100 \times \frac{R}{M} \times \frac{A}{E} \right]$$

where: N = number of coupons payable between settlement date and redemption date (maturity date, call date, put date, etc.). (If this number contains a fraction, raise it to the next whole number; for example, 2.4 = 3)
 DSC = number of days from settlement date to next coupon date
 K = summation counter

Note: The first term computes present value of the redemption amount, not including interest. The second term computes the present values for all future coupon payments. The third term computes the accrued interest agreed to be paid to the seller.

Yield (given price) with more than one coupon period to redemption:

Yield is found through an iterative search process using the "Price with more than one coupon period to redemption" formula.

Accrued interest for securities with standard coupons or interest at maturity:

$$AI = PAR \times \frac{R}{M} \times \frac{A}{E}$$

where: AI = accrued interest

PAR = par value (principal amount to be paid at maturity)

Modified duration:²

$$\text{Modified Duration} = \frac{\text{Duration}}{1 + \frac{Y}{M}}$$

2. Source for duration: Strong, Robert A., *Portfolio Construction, Management, and Protection*, South-Western College Publishing, Cincinnati, Ohio, 2000.

where *Duration* is calculated using one of the following formulas used to calculate Macaulay duration:

- For a bond price with one coupon period or less to redemption:

$$Dur = \left(1 + \frac{Y}{M}\right) \cdot \frac{Dsr \times \left[\frac{Rv + \frac{100 \times R}{M}}{\left[1 + \left(\frac{Dsr \times Y}{E \times M}\right)\right]^2} \right]}{E \times M \times Pri}$$

- For a bond price with more than one coupon period to redemption:

$$Dur = \left(1 + \frac{Y}{M}\right) \cdot \frac{\left[\frac{Rv \times \left(N-1 + \frac{Dsc}{E}\right)}{\left(1 + \frac{Y}{M}\right)^{N + \frac{Dsc}{E}}} \right] + \sum_{k=1}^N \left[\frac{100 \times \frac{R}{M} \times \left(k-1 + \frac{Dsc}{E}\right)}{\left(1 + \frac{Y}{M}\right)^{k + \frac{Dsc}{E}}} \right]}{M \times Pri}$$

Note: Bond price formulas and notations are described in this section.

Depreciation

RDV = $CST - SAL$ - accumulated depreciation

Values for DEP , RDV , CST , and SAL are rounded to the number of decimals you choose to be displayed.

In the following formulas, $FSTYR$ = $(13 - MO1) \div 12$.

Straight-line depreciation

$$\frac{CST - SAL}{LIF}$$

$$\text{First year: } \frac{CST - SAL}{LIF} \times FSTYR$$

$$\text{Last year or more: } DEP = RDV$$